CRAFT: Confidence-Aware Rubric Construction for Complex Open-Ended Evaluation with ChronoDx

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Abstract

Benchmark contamination threatens the validity of large language model (LLM) evaluation, but renewing complex benchmarks still requires costly expert annotation. Automated rubric construction may reduce this burden, yet model-generated criteria are not gold standards and remain under-validated for long, open-ended, high-stakes tasks without reference rubrics. We introduce CRAFT (Confidence-Aware Rubric-construction Agentic Framework for complex Tasks), which combines expert task definition, independent multi-model proposal, strict semantic fusion, blinded panel judging, Dawid–Skene reliability estimation, and deterministic assembly. We instantiate CRAFT in ChronoDx, an access-controlled multicenter benchmark of 120 de-identified longitudinal inpatient case–task instances across 11 specialties, covering admission diagnosis and management and discharge diagnosis and care planning. Six generators produced 12,480 proposals; fusion yielded 6,019 candidates, and complete-panel aggregation retained 2,376 atomic criteria. In a blinded three-physician audit purposively sampled from method disagreements, 90.4% of reviewed criteria were approved (allocation-calibrated: 90.3%), including 93.9% of DS-over-vote additions. Physicians preferred ensemble-retained criteria in 90.0% of matched redundancy comparisons. Across ten answer models, penalized scores ranged from 52.6 to 93.6 on a 0–100 scale, with distinct category-level profiles. CRAFT enables scalable, reliability-aware, and traceable rubric construction for complex open-ended evaluation.

1 Introduction

Benchmark contamination increasingly threatens the validity of LLM evaluation. Public benchmark items can enter training or evaluation-tuning corpora, confounding memorization with generalization (Dong et al., 2024). Because proprietary training mixtures are opaque and model versions continue to change, a fixed public test cannot reliably separate memorization from generalization indefinitely. Evaluation therefore requires renewable, exposure-conscious datasets rather than repeated scoring on one static benchmark. Domain benchmarks reveal capability structure that generic leaderboards miss (Zhao et al., 2025; Lu et al., 2026), but renewal still requires costly task curation and scoring design.

This bottleneck is most acute for complex open-ended tasks. A multiple-choice label is inexpensive to score but cannot represent a response that must integrate many facts, justify decisions, manage uncertainty, and avoid harmful omissions. A single holistic or mean-opinion score is also underdiagnostic: two answers can receive similar ratings while differing in the specific capabilities they cover, and LLM judges themselves exhibit position and self-preference biases (Zheng et al., 2023; Wang et al., 2024; Panickssery et al., 2024). Instance-specific atomic rubric criteria provide a more actionable representation, as illustrated by physician-authored criteria in HealthBench (Arora et al., 2025) and by rubric-based automatic evaluation (Liu et al., 2023; Hashemi et al., 2024). Their manual creation, however, scales poorly across long inputs, many instances, and recurring benchmark updates.

Existing work therefore leaves a construction gap: it measures increasingly complex capabilities, but does not provide a validated way to renew the instance-specific criteria needed to measure them. Automation creates a second problem rather than removing validation. A single model can omit rare but indispensable criteria, generate redundant items, or promote its own preferences. Asking another model to synthesize several lists does not show what was lost, and asking the same model family to author and approve criteria risks circular validation. Conversely, requiring experts to write

every criterion from scratch recreates the original bottleneck. A scalable construction method must therefore use automation to search broadly while exposing uncertain inclusion decisions for targeted human review. Multi-model evaluator panels offer one route to greater robustness (Verga et al., 2024), but panel disagreement still requires an explicit aggregation model rather than an unexamined vote.

We introduce CRAFT, a general framework for rapidly constructing atomic rubrics for complex tasks without assuming that any model is a gold-standard author. Diverse generators search the candidate space independently; a strict fusion agent resolves only logical equivalence; blinded judges make criterion-level decisions; and a Dawid–Skene (DS) Confidence Module estimates latent inclusion probabilities and judge-specific confusion matrices (Dawid and Skene, 1979). Deterministic assembly prevents a final language model from rewriting the selected set. The framework is task-agnostic. An upstream expert task-definition module fixes the objective, permitted evidence, output requirements, rubric schema and point ceilings, and harmful-error boundary. The remaining stages then operate on this frozen specification, structured candidates, and binary judgments.

ChronoDx is our high-stakes clinical instantiation. It uses previously non-public, access-controlled inpatient records from multiple clinical centers to reduce direct benchmark exposure, although proprietary training data prevent a guarantee of complete decontamination. Each case requires evidence integration across a temporally ordered hospitalization. The input is curated to remove direct identifiers and answer-bearing diagnosis or treatment conclusions, while retaining the clinical evidence needed to reconstruct an admission assessment or discharge plan. This setting stress-tests CRAFT; it is not the only domain to which the construction framework applies.

Our contributions are:

- CRAFT, a domain-transferable framework for fast, instance-specific rubric construction that separates proposal, fusion, judgment, aggregation, and assembly instead of treating one model as author and arbiter;
- ChronoDx, a physician-screened benchmark of 120 temporally assembled multicenter inpatient case–task instances for open-ended diagnosis, treatment/management, safety, and discharge-care reasoning, used here as a real-world stress test of the general framework; and
- a 120-case empirical study comparing six single-generator outputs, strict fusion, direct vote, complete-panel DS, and LOO-stable DS, together with a final ten-model benchmark and a blinded three-physician audit.

Figure 1 summarizes the upstream task-definition module, the five construction stages, and their ChronoDx instantiation.

2 Related Work

Domain-specific capability benchmarks. Recent benchmarks increasingly decompose professional performance rather than reporting one generic score (Zhao et al., 2025; Lu et al., 2026). However, capability decomposition usually presumes that suitable evaluation criteria already exist. CRAFT addresses this preceding construction problem: how to define the task and then derive and validate instance-specific, evidence-grounded criteria when no fixed answer key is available.

Clinical LLM evaluation. Physician evaluation of medical LLMs requires reasoning, factuality, completeness, harm, and bias rather than examination accuracy alone (Singhal et al., 2023). HealthBench uses physician-authored, conversation-specific rubrics (Arora et al., 2025); ClinicBench and IPDS/MAP cover open-ended or inpatient decision making (Liu et al., 2024; Chen et al., 2026). ChronoDx's distinctive role is to instantiate automatic case-specific rubric construction for free-text admission and discharge reasoning over a temporally assembled hospitalization.

Rubric-based and automatic evaluation. G-Eval showed that LLMs can execute explicit criteria for reference-free NLG evaluation (Liu et al., 2023). LLM-Rubric learns calibrated mappings from multidimensional rubric judgments to human ratings (Hashemi et al., 2024). SedarEval constructs a self-adaptive rubric for each question before automated grading (Fan et al., 2024), while RubricBench directly documents a gap between human-authored and model-generated rubrics (Zhou et al., 2026). These studies motivate case-specific criteria but also caution that rubric generation itself is an evaluation problem.

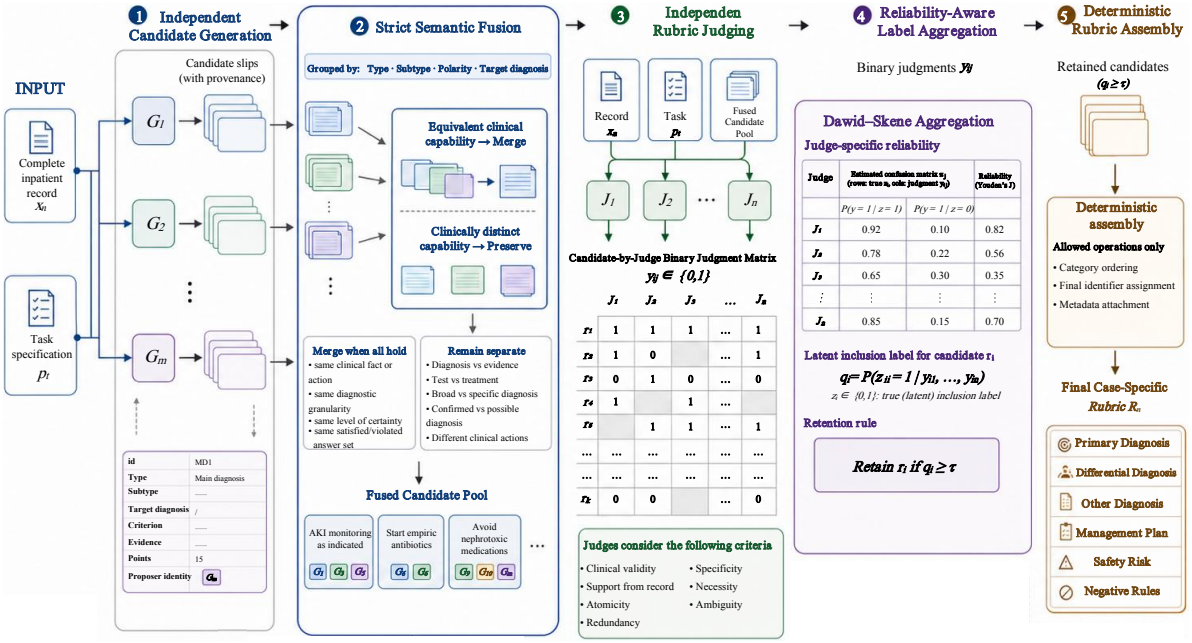


Figure 1: CRAFT lifecycle. An expert first freezes the task objective, permitted evidence, response requirements, rubric schema, point ceilings, and harmful-error boundary. Five construction stages then perform isolated candidate generation, strict equivalence fusion, blinded judging, Dawid–Skene (DS) reliability-aware aggregation, and deterministic assembly. Provenance and confidence metadata remain attached for audit.

LLM-as-a-judge and multi-judge aggregation. MT-Bench and Chatbot Arena established LLM judging as a scalable approximation to human preference while identifying position, verbosity, and self-enhancement biases (Zheng et al., 2023). Subsequent work isolated position bias (Wang et al., 2024) and linked self-recognition to self-preference (Panickssery et al., 2024). Panels of diverse evaluators can improve robustness over one large judge (Verga et al., 2024). Our use of DS differs from direct majority voting by estimating a confusion matrix for each judge and a posterior inclusion probability for every candidate (Dawid and Skene, 1979).

3 ChronoDx Instantiation and Rubric Formulation

3.1 ChronoDx Tasks

For case–task instance n, a response model receives a temporally ordered record x_n and task prompt p_{t_n}, while CRAFT independently constructs $\mathcal{R}_n$ from the same input and never from the answer being scored. ChronoDx contains 68 admission-analysis cases requiring diagnoses, differentials, diagnostic workup, treatment/management, and safety reasoning, and 52 discharge-analysis cases requiring discharge diagnoses, medication and treatment planning, instructions, warnings, follow-up, and unfinished-care planning. Thus, both tasks evaluate clinical actions as well as diagnostic conclusions.

3.2 Atomic Rubrics

Each item in $\mathcal{R}_n = \{r_{n1}, \ldots, r_{nK_n}\}$ tests one independently decidable clinical fact or action and stores its type, polarity, criterion, record-grounded evidence, and points. Positive items reward covered content; case-specific negative items penalize high-risk errors, while ordinary omission loses positive credit (Technical Supplement Sections B–C).

4 CRAFT: Confidence-Aware Agentic Rubric Construction

Figure 1 gives the complete lifecycle: expert task definition followed by five rubric-construction stages. The following subsections define each stage and its interface; the study artifacts provide the formal set definitions and production prompts.

4.1 Task Definition: The Upstream Control Module

Before candidate generation, a domain expert defines the task objective, available input, required response, rubric types and subtypes, evidence rules, point ceilings, and the boundary for case-specific harmful errors. This task adapter is frozen for

a benchmark version and supplied unchanged to generators, the fusion agent, and judges. It therefore determines what the downstream pipeline is allowed to propose, merge, retain, and score, while keeping the five downstream construction stages reusable across domains. ChronoDx instantiates the adapter separately for admission diagnosis-and-management analysis and discharge diagnosis-and-care planning; full task definitions are provided in the Technical Supplement.

This separation also supports benchmark governance: changing the objective, evidence window, scoring schema, or harmful-error boundary creates a new benchmark version rather than an undocumented prompt adjustment. The adapter therefore acts as a stable, auditable contract between domain intent and automated construction, while allowing the downstream machinery to transfer without assuming identical criteria across tasks.

4.2 Stage 1: Isolated Candidate Generation

Six generator models receive the same complete record, task prompt, schema, and negative-item boundary without seeing one another's outputs. Each proposal identifies one assessable capability with record evidence; source provenance is preserved throughout the six-generator union.

4.3 Stage 2: Strict Semantic Fusion

Proposals are blocked by type, subtype, polarity, and target diagnosis. Within a block, items merge only if they test the same fact or action at identical granularity, certainty, location, timing, and population; related or partly entailed criteria remain separate. The fusion agent may consolidate existing evidence but cannot add clinical content or select items. GPT-5.5 (reasoning effort: `xhigh`) serves as the fusion agent and assigns the fused candidate's point value subject to the fixed task-specific type and subtype ceilings; DS aggregates inclusion labels, not point values. Full task schemas, point ceilings, and production prompts are provided in Technical Supplement Sections B–D.

4.4 Stage 3: Blinded Independent Judging

Each judge receives the record, task definition, and fused candidate pool. Generator identity, proposal count, source IDs, fusion notes, original points, and other judges' decisions are hidden. For every candidate, the judge returns *include*, *exclude*, or *uncertain*, plus a rationale and confidence. The decision considers validity, evidence sufficiency, atomicity, specificity, requiredness, and redundancy. Confidence and rationale are retained for audit but do not weight DS.

The primary analysis is complete-panel: only candidates with six explicit binary include/exclude labels enter DS; abstentions are excluded. Thus every candidate in the primary pool has the same judgment panel.

4.5 Stage 4: Dawid–Skene Aggregation

Let $z_i \in \{0, 1\}$ be the unobserved inclusion label, $\rho = P(z_i = 1)$, and $\pi_{j,a,b} = P(y_{ij} = b \mid z_i = a)$ judge j's confusion matrix. Standard expectation–maximization jointly estimates the latent class prior, each judge's confusion matrix, and the candidate posterior $q_i = P(z_i = 1 \mid \mathbf{y}_i)$. A candidate is retained when $q_i \geq 0.80$. Five-of-six vote is the direct comparator; LOO refits test whether selection persists after removing each judge.

4.6 Stage 5: Deterministic Assembly

Selected candidates are ordered by clinical category and assigned final IDs. The criterion and evidence remain those emitted by strict fusion; the aggregator does not generate clinical text. Construction metadata records the posterior, vote count, and LOO status. This final stage cannot introduce or silently rewrite a criterion.

5 Experimental Setup

5.1 Data and Analysis Population

The fixed source cohort contains 120 de-identified inpatient records from multiple clinical centers: 68 admission diagnosis-and-management tasks and 52 discharge diagnosis-and-care-planning tasks across 11 specialties. Median record length is 4,955 non-whitespace characters (IQR 3,027–7,361). Only de-identified records are used in the research workflow.

Record curation and target-leakage control. Direct identifiers and answer-bearing diagnosis or treatment conclusions are removed, while evidence-bearing events are retained in hospital-time order. A physician screens every trajectory for clinical plausibility and information sufficiency.

All method totals, per-case means, and case-cluster bootstrap samples use the fixed 120-case benchmark denominator. The strict-fusion pool is defined as the 6,019 candidates with explicit binary judgments from all six judges.

Table 1: Rubric-set scale across 120 cases. Dawid–Skene (DS) denotes reliability-aware panel aggregation, and leave-one-judge-out (LOO) denotes retention after removing each judge in turn. Counts characterize construction scale, not clinical quality.

Construction method	Total	/case
Single GPT-5.5	3,113	25.9
Single Claude-Opus-4.8	1,886	15.7
Single Gemini-3.1-Pro	1,742	14.5
Single DeepSeek-V4-Pro	1,782	14.9
Single Doubao-2-Pro	1,798	15.0
Single GLM-5.2	2,159	18.0
Strict-fusion pool	6,019	50.2
Five-of-six vote	1,720	14.3
DS + LOO	1,894	15.8
Complete-panel DS	**2,376**	**19.8**

5.2 Compared Methods

The comparison includes the six individual generator outputs, the strict-fusion pool, five-of-six voting, complete-panel DS, and complete-panel DS with LOO stability. The single-model rows use exactly the same six exports supplied to fusion. Repeated generic fallback penalties are excluded from set-size counts because they are appended system rules rather than case-specific criteria. Table 1 reports the method-scale comparison.

5.3 Model Roles

The six generators are GPT-5.5, Claude-Opus-4.8, Gemini-3.1-Pro, DeepSeek-V4-Pro, Doubao-2-Pro, and GLM-5.2; the judges are DeepSeek-V4-Pro, GLM-5.2, Kimi-K2.5, Qwen3.5-397B, GPT-5.5, and Gemini-3.1-Pro. The nonidentical panels broaden evaluator-family diversity, and provenance remains hidden from judges. All calls use provider reasoning mode; identifiers and configurations appear in Technical Supplement Section E and the archived manifest.

5.4 Downstream Benchmark Scoring

A Gemini-3.1-Pro evaluator applies the 2,376 DS-retained criteria to answers from ten models. For case n, let A_n^+ be the awarded positive points, P_n^+ the sum of all available positive points, and $L_n \geq 0$ the magnitude of triggered negative penalties. The normalized score is

$$S_n = 100\frac{A_n^+ - L_n}{P_n^+}, \qquad S_n^+ = 100\frac{A_n^+}{P_n^+},$$

where S_n^+ is the positive-only counterpart. Each model's reported score is the arithmetic mean of its case-level normalized scores. Thus cases with different positive-point totals contribute equally after within-case normalization.

5.5 Expert Evaluation Protocol

The 100-decision blinded audit is allocated across 11 specialties to three physicians. Two review four specialty-specific case blocks each and one reviews three; each item is assigned to one physician. Model identity and aggregation provenance are masked. For S1–S3, physicians decide whether an individual criterion should be *included*: it must be clinically correct, record-supported, case-specific, atomic, and important enough to score.

S1 contains vote+DS criteria absent from GPT-5.5; S2 and S3 are DS-over-vote criteria with and without GPT-5.5 provenance, respectively. S4 blindly pairs an excluded redundant GPT-5.5-involved candidate with its retained representative, and the physician selects the better criterion (Technical Supplement Section F).

Generator contribution. We trace each fused candidate to its source generators and distinguish unique discovery, survival into the final set, non-exclusive final-set coverage, and equal-share fractional contribution. Table 2 reports the generator-level results.

5.6 Statistical Analysis

For each construction method we report the total and mean across 120 cases. For aggregation, we report DS–vote overlap, posterior entropy, threshold sensitivity, LOO stability, and judge-specific sensitivity and specificity relative to the inferred latent consensus. The downstream benchmark evaluation reports mean normalized overall, positive-only, and category-level scores. For the physician audit, we report inclusion count, rate, and Wilson 95% interval separately for S1–S3. The DS–vote contrast uses S2+S3, with item-level directional support, an equal-case mean, and an exact two-sided sign test over the relevant completed cases. For the fixed 80-item S1–S3 allocation, a stratum-specific beta–binomial posterior prediction with Jeffreys priors calibrates the seven unreturned decisions from one specialty block; observed counts remain explicit. S4 reports the preference rate for the ensemble-retained member of each matched redundant pair. Each item has one physician rating, so inter-rater agreement is not estimated. Allocation, masking, missing-return handling, and calibration are de-

tailed in Technical Supplement Section F. Formal DS updates and the complete robustness suite are provided in Technical Supplement Section G.

6 Results and Discussion

6.1 Construction Scale

On the fixed 120-case denominator, the six single-generator runs produced 14.5–25.9 criteria per case, corresponding to 1,742–3,113 criteria (Table 1). Strict fusion yielded 6,019 candidates with complete labels from all six judges, or 50.2 per case. Complete-panel DS retained 2,376 final criteria, or 19.8 per case. Task 1 contributed 3,735 candidates and 1,595 retained criteria, whereas Task 2 contributed 2,284 candidates and 781 retained criteria. This corresponds to 23.5 versus 15.0 retained items per fixed task denominator, consistent with the broader diagnostic, differential, management, and safety scope of admission analysis. By content group, the final set comprised 36.0% diagnosis, 46.3% management, and 17.0% safety criteria; separately, 17.0% of all items had negative polarity.

6.2 Generator Diversity and Final-Set Contribution

Across the 120 cases, the six generator exports contributed 12,480 source proposals to fusion. Within the 6,019-candidate complete six-judge pool, 4,502 candidates (74.8%) were singletons, of which 1,268 remained in the 2,376-item complete-panel DS final set.

Table 2 separates discovery volume, survival, and attribution. GPT-5.5 generated the largest singleton share (28.3%), covered 57.0% of the final set, contributed 38.7% of retained singletons, and received 31.5% of fractional credit. Claude-Opus-4.8 contributed a further 21.8% of retained singletons. Claude-Opus-4.8 proposed fewer items but had the highest proposal success (55.7%) and the highest complete-panel singleton retention (54.1%, versus 38.6% for GPT-5.5). The other four generators each contributed 10.9%–13.9% of fractional credit, confirming that final coverage did not collapse to one generator's proposal distribution.

6.3 Primary Expert Comparison

The three-physician audit targets disagreements among DS, five-of-six vote, and GPT-5.5. Judgments were returned for 73 of 80 S1–S3 items: 20/24 S1, 23/26 S2, and 23/23 S3 items were approved (83.3%, 88.5%, and 100.0%).

Among criteria rejected by vote but retained by DS, physicians approved 46/49 (93.9%; 95% CI, 83.5–97.9); the calibrated estimate over the allocated S2+S3 items was 93.5%. GPT-5.5 proposed no S3 item, indicating that this gain was not simply its provenance effect. All ten completed case blocks had a majority of their S2+S3 items approved; the equal-block mean was 94.9%, and an exact two-sided sign test gave $p = 0.0020$. The result therefore reflects consistent support across the audited blocks rather than a small number of high-yield cases. Full stratum analyses appear in the Technical Supplement.

In S1, physicians approved 20/24 criteria absent from GPT-5.5's proposals, showing that generator diversity also adds valid coverage. Overall, 66/73 returned S1–S3 decisions were positive (90.4%); allocation calibration gave 90.3%. These rates describe targeted disagreements, not full-pool precision. The seven unreturned decisions bound the full-allocation rate between 82.5% if all were negative and 91.3% if all were positive. This range avoids a uniformly favorable imputation while showing that the interpretation is not tied to one missing-rating assumption. Further sensitivity results appear in the Technical Supplement.

S4 addresses the main reason that a GPT-5.5 proposal can be absent from the final set: a retained criterion already covers the same capability. In the blinded paired comparison, physicians preferred the ensemble-retained criterion in 18 of 20 matched pairs (90.0%). This result supports the fusion and aggregation pipeline's choice of representative criterion rather than interpreting redundant GPT-5.5 exclusions as lost clinical content.

6.4 Vote Versus DS and Robustness

Within the 6,019-candidate strict-fusion pool, the five-of-six rule retained 1,720 and DS at $\tau = 0.80$ retained 2,376. Every five-of-six selection was also selected by DS; DS additionally selected 656 four-to-two vote items, giving a Jaccard similarity of 0.724. This difference reflects judge-reliability weighting. The physician audit supports the direction of this expansion: 46/49 reviewed S2+S3 items were approved, although the purposive sample does not estimate the full-pool precision of all 656 additions.

The conservative DS+LOO subset contained 1,894 items (79.7% of the primary DS set). The aggregation diagnostics below show how heterogeneous judge profiles allow a fixed vote and

Table 2: Generator provenance in the 6,019-candidate complete-panel pool. A singleton has one source proposal; proposal success is the fraction mapped to a retained candidate. Final-set coverage is non-exclusive, whereas fractional contribution divides shared criteria equally and sums to 100%. These are descriptive construction measures; larger values do not uniformly imply better generator quality.

Generator	Raw props.	Panel singletons	Pool share	Retained singletons	Retained share	Proposal success	Final coverage	Fractional credit
GPT-5.5	3,113	1,273	28.3%	491	38.7%	43.5%	57.0%	31.5%
Claude-Opus-4.8	1,886	510	11.3%	276	21.8%	55.7%	44.4%	21.1%
Gemini-3.1-Pro	1,742	620	13.8%	121	9.5%	41.2%	30.2%	11.6%
Doubao-2-Pro	1,798	684	15.2%	117	9.2%	37.3%	28.6%	10.9%
DeepSeek-V4-Pro	1,782	676	15.0%	120	9.5%	37.1%	28.2%	11.0%
GLM-5.2	2,159	739	16.4%	143	11.3%	38.3%	34.7%	13.9%

Table 3: Dawid–Skene (DS)-inferred judge profiles relative to latent panel consensus. Balanced reliability is the mean of sensitivity and specificity. These are aggregation parameters, not external clinical accuracy. Best, second-best, and third-best values are shown in red, blue, and bold, respectively.

Judge	Sensitivity	Specificity	Balanced
DeepSeek-V4-Pro	**0.865**	**0.840**	**0.853**
GLM-5.2	0.826	**0.893**	**0.859**
Kimi-K2.5	**0.950**	0.622	0.786
Qwen3.5-397B	**0.895**	0.790	**0.842**
GPT-5.5	0.608	0.838	0.723
Gemini-3.1-Pro	0.767	**0.850**	0.808

reliability-weighted DS to disagree.

6.5 Composition Across Selection Rules

Selection changed coverage more than construct composition. Vote retained 1,720 items versus 2,376 for DS, yet their diagnosis, management, safety, and negative-item shares differed by at most 1.2 percentage points. Specifically, the DS set comprised 36.0% diagnosis, 46.3% management, and 17.0% safety criteria, with 17.0% negative polarity. DS+LOO retained 1,894 items, with every major share within 0.9 points of DS. Thus, the expansion primarily increased breadth rather than redefining the clinical construct. We use DS for broader physician-supported coverage and expose LOO stability as audit metadata; full compositions are reported in the Technical Supplement.

6.6 Aggregation Diagnostics

Judge profiles were asymmetric (Table 3): Kimi-K2.5 favored sensitivity (0.950) over specificity (0.622), whereas GLM-5.2 had the highest balanced reliability (0.859). Task-specific estimates differed from the global fit by at most 0.017.

Posterior mass was sharply separated (86.5% below 0.05 or above 0.95). DS retained every five- or six-vote item and added only the 656 four-to-two items. Selection was unchanged for $\tau = 0.60$–0.80; leave-one-judge-out refits gave Jaccard similarity 0.827–0.914 and a 1,894-item stable subset. Full threshold, entropy, and residual diagnostics appear in the Technical Supplement.

6.7 Final Ten-Model Benchmark Results

All ten answer models received Gemini-3.1-Pro-based rubric scores on the same benchmark case set. Table 4 reports positive-only, penalized, and category-level scores on a 0–100 scale.

Mean penalized scores ranged from 52.60 to 93.60 points. DeepSeek-V4-Pro and Doubao-2-Pro incurred the two largest reductions from case-specific negative criteria (-7.66 and -7.04, respectively), whereas GPT-5.4 lost only 0.18 points. On this benchmark, the larger penalties signal that the former two models more frequently produced clinically incorrect or potentially harmful judgments, suggesting greater safety risk if their outputs are used without expert oversight. Category scores further exposed distinct profiles: Doubao-2-Pro scored 82.28 on follow-up but 54.90 on safety, while Qwen3.5-30B scored 77.94 on follow-up but 46.34 on primary diagnosis. DeepSeek-V4-Pro also combined strong primary diagnosis (86.17) with lower differential diagnosis (61.29) and safety (70.89), illustrating why atomic profiles matter.

The results suggest distinct model niches rather than a simple access- or scale-based ordering. Closed-weight models occupied the top three positions but varied substantially within the group. Open-weight DeepSeek-V4-Pro approached the middle closed-weight tier in positive coverage and was strong on primary diagnosis, yet incurred the largest penalty. Other open-weight systems scored lower overall but remained competitive on selected

Table 4: Final ChronoDx results for ten models on 120 cases, grouped by closed- versus open-weight access and then by model family. Pos./Pen. are positive-only/penalized scores on a 0–100 scale, and Δ is the penalty-induced reduction in score points. Up/down arrows indicate whether higher/lower values are better. Category columns are positive-only: PD, DD, and OD denote primary, differential, and other diagnosis; MP, DM, NT, and FU denote management plan, discharge instruction, precaution, and follow-up; Safety denotes safety-risk identification. Best, second-best, and third-best results are shown in red, blue, and bold, respectively.

Model information		**Overall score**			**Clinical capability profile**							
Answer model	**Weights**	**Pos.↑**	**Pen.↑**	**Δ ↓**	**PD↑**	**DD↑**	**OD↑**	**MP↑**	**Safety↑**	**DM↑**	**NT↑**	**FU↑**
Closed-weight models												
GPT-5.5	Closed	**93.90**	**93.60**	**0.30**	**95.15**	**86.41**	**96.11**	**95.20**	**92.57**	**96.83**	**83.66**	**94.79**
GPT-5.4	Closed	**76.23**	**76.05**	**0.18**	77.43	**71.52**	79.29	76.45	**79.78**	**89.82**	**82.14**	**87.52**
Claude-Opus-4.8	Closed	**85.91**	**83.91**	2.00	**91.26**	**72.06**	**94.22**	**87.27**	**88.14**	**86.02**	**66.27**	**86.57**
Claude-Opus-4.6	Closed	68.95	66.68	2.27	76.70	52.48	72.79	67.68	70.23	77.88	71.58	81.34
Gemini-3.1-Pro	Closed	66.89	65.77	**1.12**	72.72	54.13	73.75	66.55	72.81	77.48	64.40	82.61
Doubao-2-Pro	Closed	70.01	62.97	7.04	63.50	63.98	73.96	77.45	54.90	69.90	66.01	82.28
Qwen3.5-Plus	Closed	63.86	60.97	2.89	62.04	54.49	70.32	65.13	68.34	74.11	69.67	78.57
Open-weight models												
DeepSeek-V4-Pro	Open	75.76	68.10	7.66	**86.17**	61.29	**79.81**	**78.13**	70.89	75.46	68.69	77.19
DeepSeek-V3.2	Open	57.21	53.62	3.59	64.55	44.00	59.25	60.17	50.73	70.70	67.73	65.67
Qwen3.5-30B	Open	56.62	52.60	4.02	46.34	46.94	64.24	60.82	53.96	73.74	69.78	77.94

discharge-care dimensions. Because several proprietary parameter counts are unavailable and architectures and reasoning budgets differ, these are ecosystem-level observations rather than causal effects of access type or model scale.

Negative criteria also changed ranking: Doubao-2-Pro's 7.04-point penalty moved it behind Claude-Opus-4.6 and Gemini-3.1-Pro, while DeepSeek-V4-Pro retained fourth place despite the largest reduction (7.66). Broad content coverage can therefore coexist with consequential clinical errors that warrant caution in use.

6.8 Interpretation

CRAFT separates search, equivalence resolution, suitability judgment, aggregation, and assembly. This prevents a synthesis model from silently rewriting the selected set and keeps every item traceable to evidence, provenance, judgments, posterior, and LOO behavior.

No generator dominated both coverage and selectivity: GPT-5.5 contributed the broadest coverage, Claude-Opus-4.8 had higher proposal success, and each other generator contributed 10.9%–13.9% of fractional credit. Physician-approved S1 items confirm clinically valid coverage beyond GPT-5.5 alone.

DS preserved every five- or six-vote item and added the four-to-two stratum; physicians approved 93.9% of reviewed additions. This supports the audited boundary without implying that all 656 additions are correct; LOO provides a stricter alternative when stability is preferred.

Physicians preferred the retained representative in 90.0% of redundancy pairs, showing that exclusion can reflect successful fusion rather than lost content. Together, the analyses expose construction and scoring dependencies for audit without treating model consensus as clinical ground truth.

7 Conclusion

Renewing contamination-prone benchmarks requires scalable instance-specific criteria without treating any generator as ground truth. We introduced CRAFT, which separates independent proposal, strict equivalence fusion, blinded judging, reliability-aware aggregation, and deterministic assembly while preserving evidence and provenance. Across 120 ChronoDx cases, CRAFT retained 2,376 of 6,019 fused candidates. Physicians endorsed 93.9% of the purposively sampled DS-over-vote additions and preferred the retained criterion in 90.0% of matched redundancy comparisons. The resulting rubrics distinguished ten answer models, with mean penalized scores spanning 52.6–93.6 points on a 0–100 scale. Capability profiles and harmful-error penalties further showed that broad coverage need not imply clinical safety. Together, expert task definition, targeted review, and traceable aggregation position CRAFT as an auditable construction recipe for other long-form, high-stakes domains.

8 Limitations

This study establishes CRAFT on 120 cases from multiple clinical centers and two inpatient tasks. Evaluation on additional institutions, specialties, and task families will further test portability. The physician audit intentionally focuses on informative method-disagreement items, with one physician rating each assigned item; future benchmark refreshes can add overlapping multi-rater review and prospective samples. Finally, aggregation and downstream scores depend on the selected model panel and evaluator, making periodic revalidation with new model and prompt versions a useful part of benchmark maintenance.

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